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Palaeomagnetic study of Archaean Banded Hematite Jasper Rocks from the Singhbhum–Orissa Craton, India

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Abstract

The Banded Hematite Jasper Formation within the Iron Ore Supergroup of the Singhbhum Craton in eastern India comprises fine alternating layers of jasper and specularite. It was deposited at ~3000 Ma and deformed by a mobile episode at ~2700 Ma. Hematite pigment (<1 μm) mixed with cryptocrystalline silica and specularite (>10 μm) is chiefly responsible for red to brown rhythmic bands in the hematite jasper facies although thermomagnetic study also shows that minor amounts (1–2%) of magnetite are present. Palaeomagnetic study identifies a dual polarity remanence resident in hematite (D/I = 283/60°, α_{95} = 12°) which predates deformation. Studies of the fabric of magnetic susceptibility and rock magnetic results suggest a diagenetic origin for this magnetisation with the hematite formed from oxidation of primary magnetite. The palaeopole (32°E, 24°N, dp/dm = 14/18°) records the earliest post-metamorphic magnetisation event in the Orissa Craton. A minimum apparent polar wander motion of the Orissa–Singhbhum craton of through 80° is identified during Late Archaean times (2900–2600 Ma).

Keywords: Banded hematite jasper; Archaean; Palaeomagnetism; India

1. Introduction

Banded hematite jasper (BHJ) facies occur widely in the Late Archaean–Palaeoproterozoic ironstone formations and are potentially attractive subjects for palaeomagnetic study because they usually possess strong and stable remanence characteristics. The Early Archaean examples tend to have a ferromagnetic content dominated by magnetite whilst the Late Archaean and Palaeoproterozoic examples are predominantly hematite-bearing (Hutchinson, 1981). Magnetite in BHJ facies may either be (i) detrital and have survived deposition in a neutral or weakly oxidising environment (in which case it holds the prospect of recovering remanence of detrital (DRM) or post-depositional detrital (PDDRM) origin), or (ii) it may be a diagenetic constituent formed in mildly reducing and alkaline conditions (Piper, 1987). Hematite occurs as large grains of both specularite (of detrital or authigenic origin) and as pigment. It is likely to be produced in part by oxidation of precursor magnetite and thus possess a chemical remanent magnetisation (CRM). The age of alteration leading to oxidation is then a crucial

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question to be addressed when dating the magnetisation.

The main limitations associated with study of these rocks are twofold. Firstly the complex paragenesis of the two magnetic minerals cannot typically be dated, although in these fine grained siliceous rocks it is likely to predate lithification. Formation can sometimes be constrained with respect to a later episode of folding by application of the palaeomagnetic fold test. Secondly hematite has a strong crystalline anisotropy whilst both the magnetite and the hematite may possess a shape fabric (Tarling and Hrouda, 1993). As a consequence, it is anticipated that depositional and/or later strain-induced fabrics might influence the remanence directions. The magnetic fabric may, however, also be of value because it can be compared with bedding to recognise any relationship with the palaeohorizontal. If the maximum and intermediate susceptibility directions lie in the plane of bedding and the minimum susceptibility is perpendicular to bedding a possible depositional origin is identified; if the axes are linked to tectonic trends a deformational origin is recognised (Tauxe et al., 1990).

2. Geological framework

The Singhbhum–Orissa Craton extends over an area of approximately 25,000 km^2 in NE India. The Iron Ore Supergroup (IOG) is a folded but only weakly metamorphosed ca. 350 m thick succession deposited on the metamorphic basement. The outcrops are concentrated mostly within two narrow NE–SW-trending basins (Fig. 1). BHJ rocks occur as a stratigraphic member of the Banded Ironstone Formation, a distinctive marker horizon in the IOG. The Orissa BIF Formation is considered (Majumdar and Chakraborty, 1979) to have been deposited as a magnetite-bearing silica gel in which magnetite was present both as a fine dust in silica and as large grains forming indistinct Fe-oxide bands. Thus the physical form of the precursors of iron oxide are likely to have been crystalline magnetite admixed with silica and water, and a homogeneous mixture of iron and silica in a colloidal state which produced the jasper. The fine banding suggests a tidal motivation to BHJ sedimentation and possible lunar cycles have been reported by a number of workers (e.g. Mohr, 1975).

The Singhbhum–Orissa Craton (Fig. 1, inset) is bounded by the arcuate Copper Belt Thrust in the north and the Sukinda Thrust in the south (Saha et al., 1988). A long Precambrian history recorded within this crustal unit includes three distinct orogenic episodes accompanied by granite emplacement at 3200, 2700 and 850 Ma (Sarkar et al., 1969, 1979). The first episode, incorporating crustal growth and thickening, was followed by several episodes of sedimentation and injection by mafic magmas. Subsequent deformation was responsible for development of large NE–SW trending basins in which deposition of the Banded Ironstone Formations occurred later in mid-Archaean times at about 3000 Ma (Sarkar and Saha, 1977).

Supracrustal rocks underlying the IOG comprise a clastic succession incorporating lithic wackes, shales, lavas and tuffs. The IOG is overlain by a manganiferous shaley horizon. Rb–Sr dating of the underlying metamorphic basement suggests that uplift and cooling had occurred by ~3150 Ma whilst the Singhbhum Granite emplaced into IOG rocks is dated ~2900 Ma (Sarkar et al., 1969, 1979; Saha and Ray, 1984). These constraints suggest a depositional age of about 3000 Ma for the IOG. It is influenced by two later deformations which produced fold generations along NNE–SSW and WNW–ESE trends; the former is the more prominent (Sen, 1982; Saha, 1988). The main regional structure influencing the IOG is a NNE trending low plunging synclinorium overturned towards the east and with a U-shaped surface configuration (Chakraborty and Majumdar, 1986). The BHJ rocks of this study come from local folds superimposed onto the eastern limb and hinge area of the large folded structure (Fig. 1).

3. Field sampling and laboratory methods

Samples were collected as blocks at fresh outcrops distributed at the 13 locations shown in Fig. 1 and oriented by magnetic compass; a simple gradiometer was used to recognise that no local magnetic deflections due to the rock outcrop were present. The sample locations and the codes used in the following discussion are Noamundi (N, 2 sites), Thakurani (TH, 2 sites), Bolani (B, 1 site), Kiriburu (KB, 1 site), Joda (J, 2 sites), Jiling (JL, 1 site), Khandabandh (KD, 2 sites) and Khandadhar (K, 2 sites).

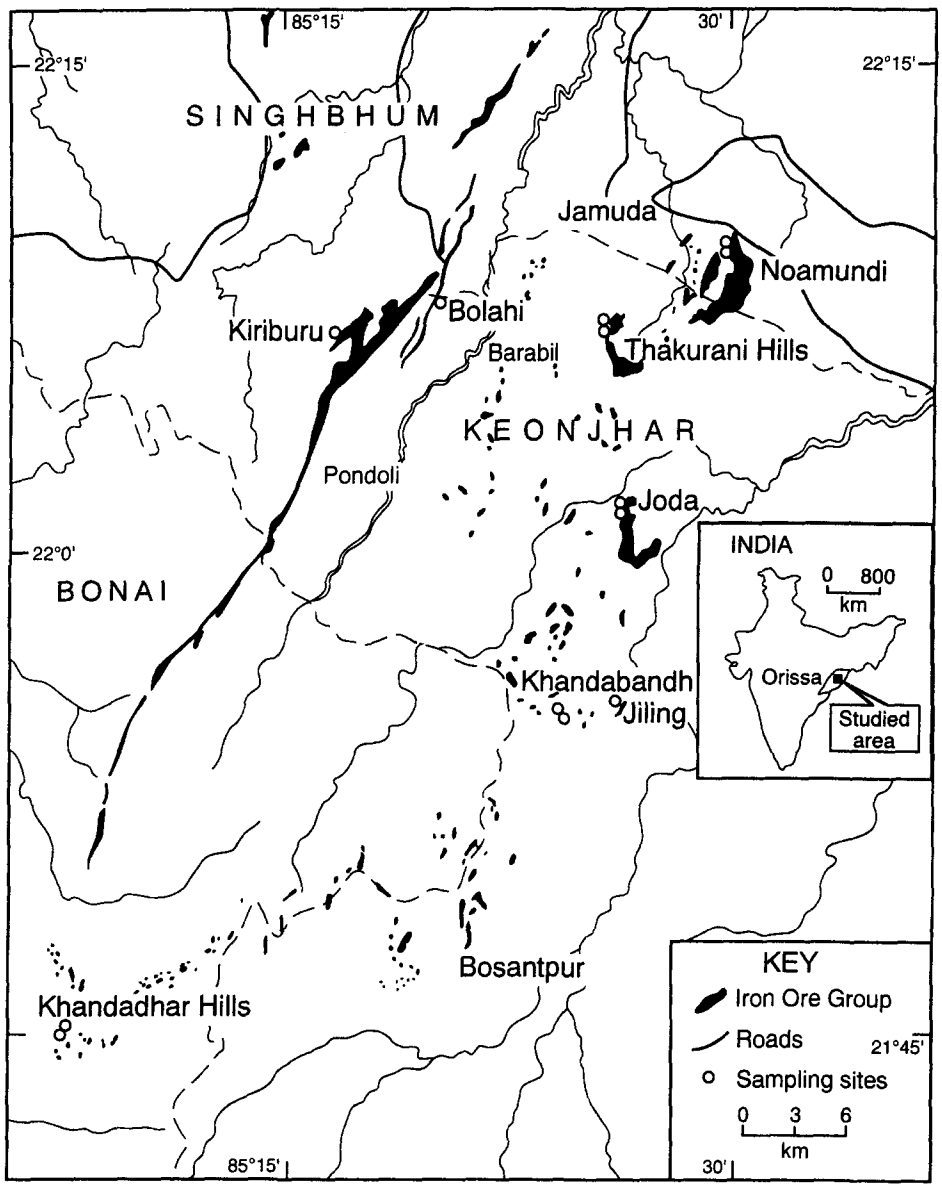


Fig. 1. Map of part of the Orissa–Singhbhum Craton in northeast India showing the outcrop (black) of the Iron Ore Group. The locations sampled in the present study are shown together with locations named in the text. The inset map shows the regional location.

In the laboratory 2.5-cm-diameter cores were drilled from the blocks and 2.2-cm cylinders cut from these cores. Initial susceptibilities were measured and anisotropies of susceptibility were then determined for each core using a 'Minisep' delineator. Subsequent thermal demagnetisation was undertaken in steps of 100 or 50°C to 600°C and then in steps of 30°C to 690 or 720°C. Demagnetisation and measurement utilised spinner magnetometers and Magnetic Measurements thermal demagnetisers. The apparatus is contained within a large set of Rubens coils in which the residual field is continuously monitored to minimise secondary remanence acquisition. The components comprising the total natural remanent magnetisations (NRMs) were recognised by study of orthogonal projections of the

remanence vector; components were then isolated interactively and their directions calculated by principal component analysis. Small chips of the cores were used for polished section study by reflected light and for thermomagnetic experiments using a computer-controlled horizontal Curie balance.

4. Fabrics of susceptibility

Initial susceptibilities ranged from 4 to 83 × 10^{-4} S.I. units. The anisotropy of magnetic susceptibility (AMS) is conventionally expressed in terms of a triaxial ellipsoid with maximum (k_1), intermediate (k_2) and minimum (k_3) susceptibility values. Anisotropy magnitudes (k_1/k_3) are typically only a few percent in the BHJ facies and a plot of lineation parameter L against foliation parameter F shows that these weak fabrics tend to be oblate (Fig. 2). This is not exclusively the case but when samples have oblate fabrics they are more strongly oblate than is the case for the prolate samples. This could suggest that magnetic fabrics in these rocks are attributable to alignment of the basal planes of hematite grains (Lee et al., 1990). The q value is defined by: $q = 2(k_1 - k_2)/(k_1 + k_2 - 2k_3)$.

In the Orissa BIF samples of this study values range from 0.257 to 0.675. In young sediments values of $q < 0.7$ point towards weaker oblate fabrics controlled by deposition (Hamilton and Rees, 1971) and contrast with larger values in tectonic fabrics; hence these low values suggest a depositional origin.

Whilst a primary depositional control is also suggested by a tendency for the k_1 and k_2 axes to lie close to the plane of the bedding and for k_3 axes to be perpendicular to this plane (Fig. 3), this effect could be produced by the fine laminated bedding and cannot confidently be used to isolate the origin of the ferromagnetic components. This relationship to bedding becomes less pronounced in sites from the south and east of the study area where the k_1 and k_2 axes depart considerably from the local bedding planes and suggest a tectonic influence on the fabrics. This is indicated by a weak lineation in the magnetic fabric and is illustrated by a tendency of the k_1 axes to be aligned with the NNE–SSW trend of the basins (cf. Figs. 1 and 3). This direction corresponds to the predominant trend of the major folds affecting the BHJ rocks.

5. Rock magnetism

Thermomagnetic studies (saturation remanence, J_s, versus temperature) of BHJ rocks identify hematite in both the specularite and jasper bands.

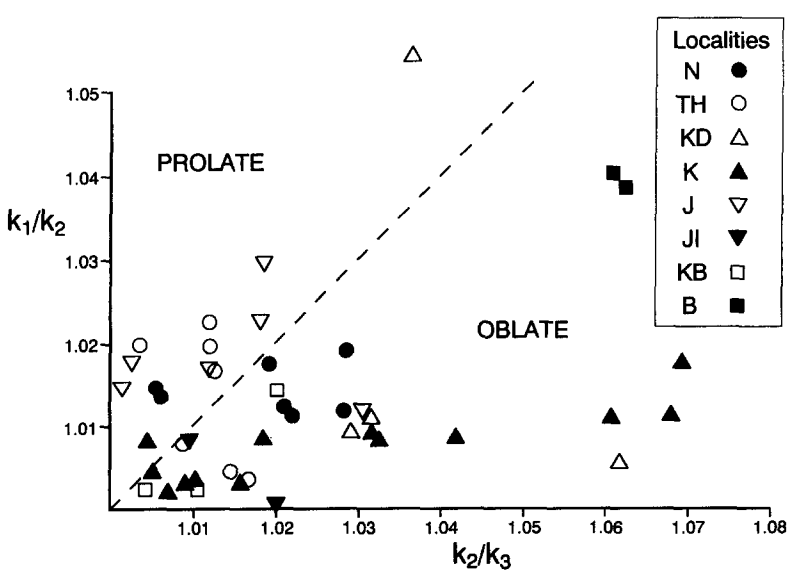


Fig. 2. The fabric of susceptibility in the Banded Hematite Jasper rocks shown as a plot of the L (lineation) parameter (k_1/k_2) plotted against the F (foliation) parameter (k_2/k_3).

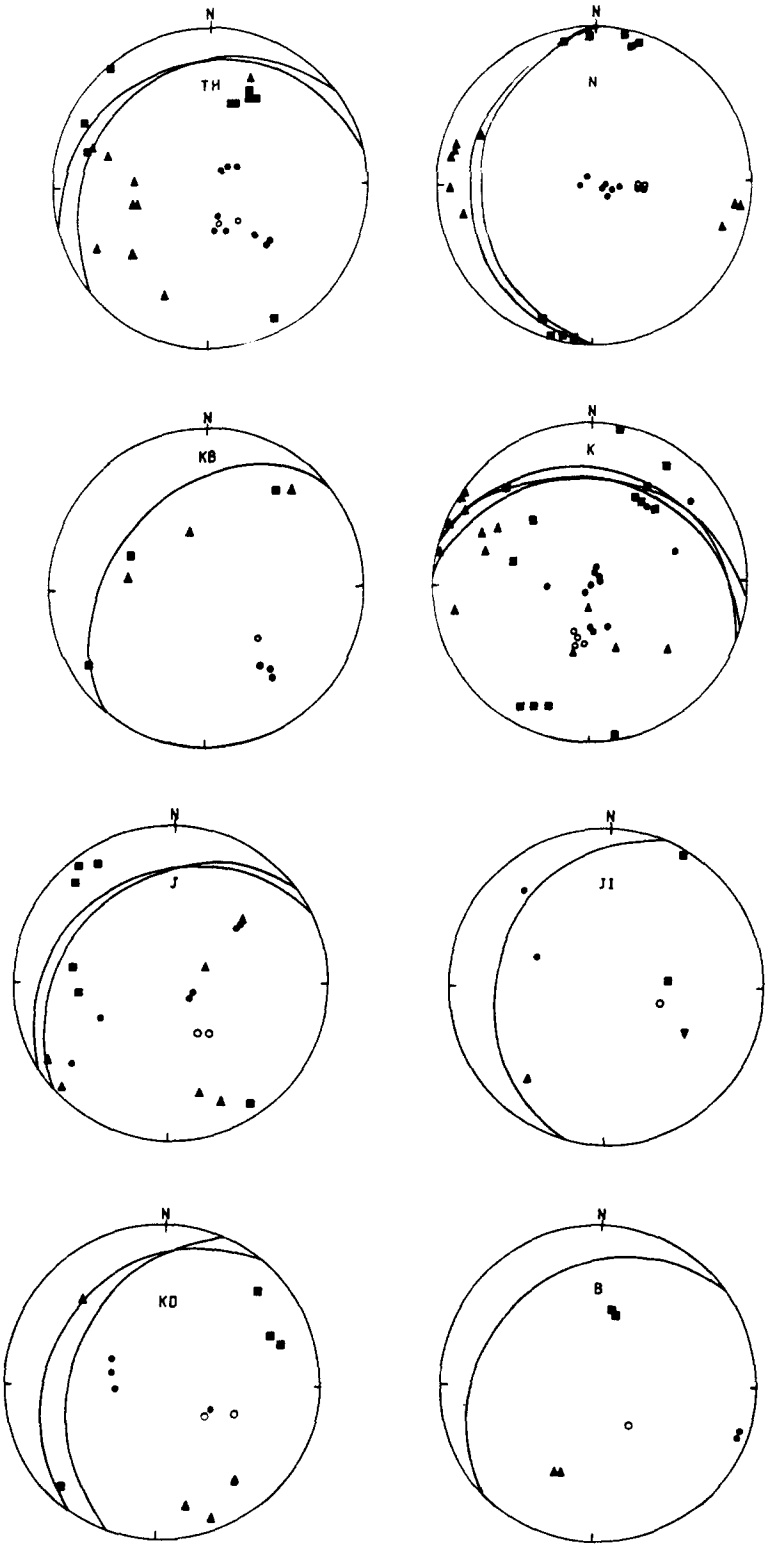


Fig. 3. Lower hemisphere plots of the directions of the maximum (k_1, squares), intermediate (k_2, triangles) and minimum (k_3, circles) axes of susceptibility at the eight sampled districts shown with respect to the orientation of the local bedding (arcs).

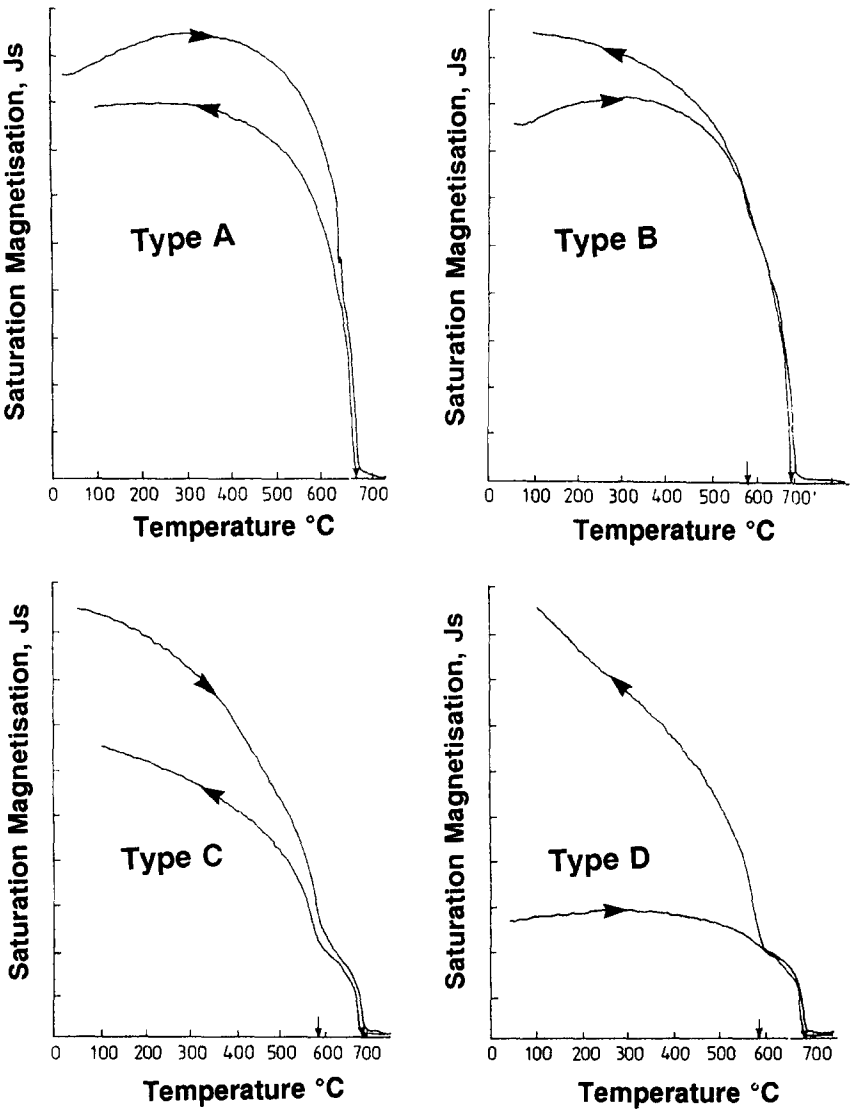


Fig. 4. Representative thermomagnetic curves (saturation magnetisation J_s, versus temperature in °C) from the BHJ facies. The saturation magnetisation values are arbitrary and the arrows indicate heating and cooling curves.

Magnetite and hematite are distinguished with this technique both by contrasting Curie points and because the latter mineral has a saturation remanence only about 2% of the former. In the BHJ facies the saturation remanence (J_s) is little affected by heating in air, whereas some increases in J_s during cooling in the jasper bands indicate that reduction has taken place to produce (at the most) a few percent of magnetite (Fig. 4). Small amounts of primary magnetite are also recognised by low temperature susceptibility study (Mallik et al., 1993) and by thermomagnetic study of the jasper (Fig. 4). Thermomagnetic behaviours are of four types; they occur in roughly equal numbers and are classed as A to D in Fig. 4. The single Curie points at 680°C identify the exclusive presence of hematite (types A, B and D). This undergoes little change with heating although a slight decrease in coercive force of unsaturated hematite appears to gradually align grains with higher coercive force with the applied field

and account for the small initial rise in J_s on the heating curve. Double Curie points at ca. 580°C and 680–690°C in both heating and cooling cycles (type C) indicate both magnetite and hematite. Because the saturation magnetisation of magnetite is about 50 times greater than hematite, only a small amount of magnetite (volumetrically less than the hematite in the examples shown) is required to produce this kind of curve; a faint Curie point between 580 and 600°C, as shown by type B in Fig. 4, is equivalent to only about one part per thousand of magnetite. An increase in J_s on cooling (type D) is the signature of small additional amounts of magnetite which might be formed by conversion of paramagnetic minerals or by reduction of the hematite.

The thermomagnetic curve types showing two distinct Curie points at 580–620°C and 680–690°C are of two types. In type C there is a marked reduction of J_s on cooling; it is then probable that the magnetite is a cation-deficient maghemite formed during diagenesis which breaks down into hematite and magnetite on cooling. A marked increase in J_s on the cooling curve is a possible signature of magnetite formation by reduction of hematite (type D in Fig. 4). The pigmentary hematite is found to convert to magnetite between 600 and 700°C (see also Mallik et al., 1993); as the thermomagnetic experiments on the BHJ samples were conducted in air, it would appear that a reducing agent within the rock contributes to this reaction. This might be an organic constituent linked to spherulitic structures found in the jasper bands although these probably have little role to play in the reduction of hematite because they are inactivated by oxidation below 400°C. It is possible that the presence of water in the jasper bands may reduce the f_{O_2} sufficiently for hematite to reduce to magnetite at temperatures in excess of 600°C (Johnson et al., 1972). The greater surface area of the fine grained pigment may also accentuate the reduction process.

6. Palaeomagnetism

Typical orthogonal projections of the thermal demagnetisation results are shown in Fig. 5. Rarely the remanence is dominated by only one component (e.g. Bolani site 2 and Kiriburu site 1 in Fig. 5); more commonly (and often at the same site) a thermally distributed lower blocking temperature component is unblocked between 100 and 550 or 600°C. The presence of magnetite is clear in the Curie curves of all four types, and the thermally distributed nature of the magnetite component is also reflected in the thermal decay curves. This component is usually subtracted to isolate a discrete high blocking temperature component between 600 and 690°C or 720°C (e.g. Thakurani, Khandadhar, Joda and Bolani site 1 examples in Fig. 5). At Thakurani 3 component behaviour includes a distributed NNE component which is subtracted by the Curie point of magnetite to isolate a steeper NE component. At other sites illustrating 2 component behaviour a lower blocking temperature component (for example NW shallow at Bolani site 1, NNE- at Khandadhar in Fig. 5) is progressively subtracted to the Curie point of magnetite. Other sites are dominated by a single component with narrow (Bolani site 2) to distributed (Kiriburu site 1) behaviours. Where the hematite component is discrete it tends to have a narrow unblocking temperature spectrum and is sometimes subtracted by a single 30°C step between 600 and 690°C or 720°C (e.g. Bolani site 1 and Khandadhar site 1 in Fig. 5).

The components which are fully subtracted by the Curie point of magnetite are near-random at most sites and were evidently acquired at various times during the long post-Archaean history of this terrane. Coherent groups are however, observed at sites TH1, N2, and J2 (Table 1). At site N2 the component subtracted by the Curie point of magnetite is displaced only marginally from the component resolved above this and it seems likely that both minerals carry remanence of the same origin. The hematite-held components have predominantly westerly declinations and shallow to steep inclinations with near-antiparallel components present at two sites (TH1 and J2). To assess the age of remanence with respect to folding, the plunge is first corrected for; the residual apparent dip is then rotated to the palaeohorizontal (Table 1). As a result the inclination distribution is reduced and E-directed components at sites J1 and B1 move to group with the two E-sites suggesting an ancient field direction of opposite polarity (Fig. 6).

The precision of the group mean of 8 component directions (D/I = 283/60°) improves from k = 14.3

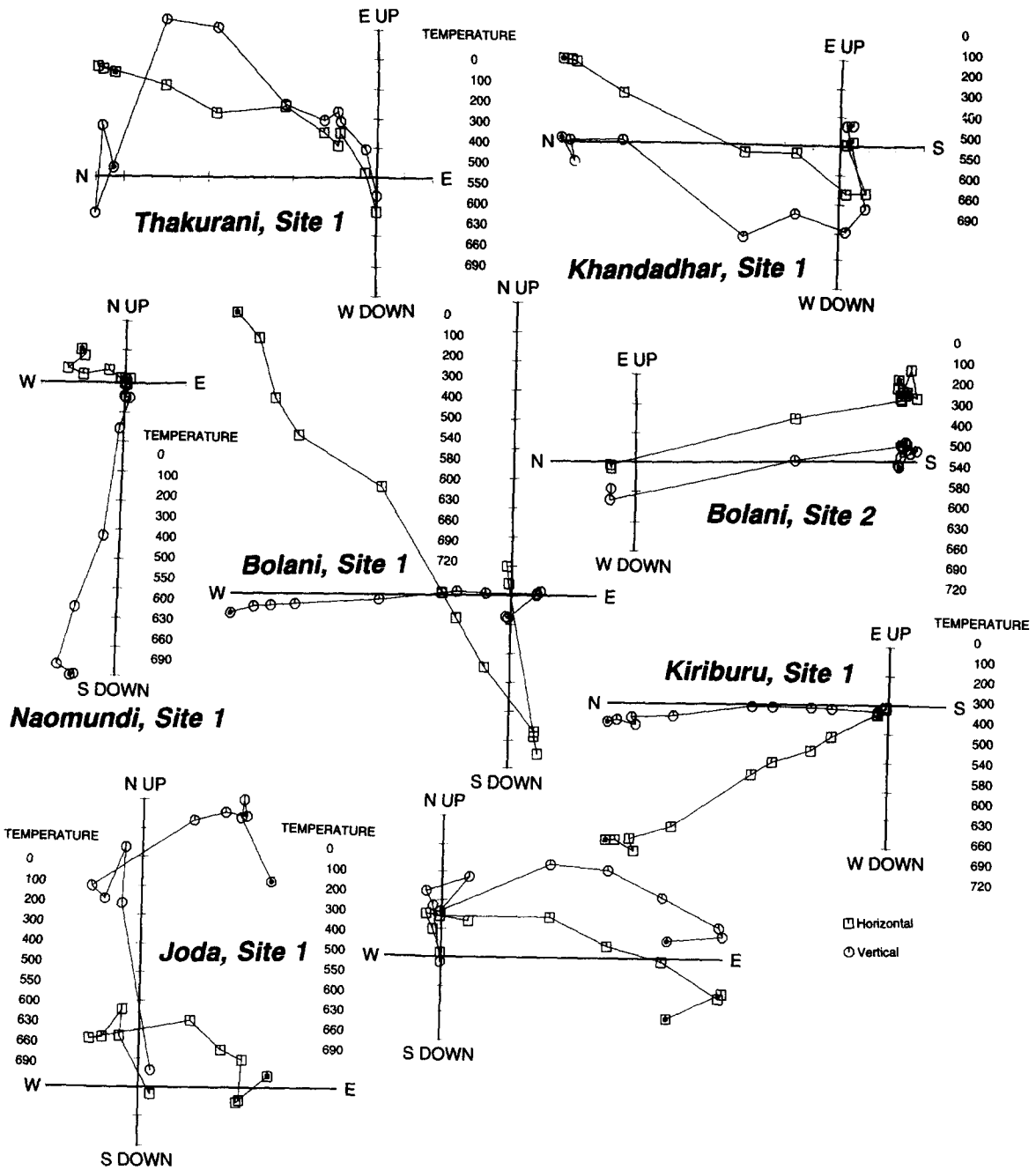


Fig. 5. Orthogonal projections illustrating some examples of progressive thermal demagnetisation of hematite jasper rocks from the Singhbhum–Orissa Craton. Projections onto the horizontal plane are shown as squares and projections onto the vertical plane as circles; the temperature steps are listed in °C.

to $k = 21.7$. This suggests a prefolding origin but is not sufficient to exclude the possibility of post-folding origin (McFadden, 1990). Of the eight sites comprising this population three are reversed with respect to the remainder. The angle $\gamma_c = 28°$ does not require rejection of the hypothesis of a common

Table 1
Site mean palaeomagnetic results from Banded Hematite Jasper facies, Singhbhum–Orissa Craton

Site code	N/n	(in situ)		R	k	α_{95}	Plunge of fold	Bed dip	(tilt adj.)	
		D	I						D	I
TH1 (m)	4/4	9.0	−5.6	4.66	12	23.3	45/32	30/345	7.6	0.8*
TH1 (h)	4/4	66.4	−51.6	3.83	18	22.3	45/32	30/345	59.2	−50.0
TH2 (h)	4/4	252.6	11.8	3.93	45	13.8	45/32	23/320	254.8	10.9*
N1 (h)	5/4	284.4	78.1	4.00	221	2.0	10/15	20/272	281.0	57.2
N2 (m)	4/4	224.9	79.4	3.93	46	13.7	10/15	26/275	266.8	55.9
N2 (h)	4/4	245.6	80.7	3.93	18	22.4	10/15	26/275	273.2	55.0
K1 (h)	4/4	321.0	6.2	3.20	30	17.0	32/350	30/15	321.0	5.6*
K2 (h)	4/3	274.4	44.9	2.96	46	18.4	32/350	30/15	275.3	45.0
KB1 (h)	3/3	349.6	9.8	2.89	19	29.3	40/22	38/313	+354.1	−6.3
B1 (h)	3/3	168.0	−1.5	2.93	29	23.5	40/206	26/320	+159.7	18.3
J11 (h)	2/2	343.2	76.4	1.99	138	21.4	20/333	30/287	318.2	64.4
J1 (h)	4/4	343.8	−62.5	3.99	203	6.5	20/125	34/320	132.9	−61.3
J2 (m)	4/2	264.3	−32.1	1.99	163	–	20/125	30/30	262.5	−9.3*
J2 (h)	2/2	89.2	−59.8	1.99	34	–	20/125	30/30	155.2	−67.2
Group mean results:										
8 components, in situ:										
		255.6	72.4	7.51	14	15.2				
8 components, tilt adjusted:										
		282.7	60.2	7.68	22	12.2				
Palaeomagnetic pole: 32.1°E, 23.8°N (dp/dm = 14.0/18.4°)										
'B' magnetisations, 4 components:										
		347.0	3.1	3.81	15	24.1				
Palaeomagnetic pole: 194.6°E, 36.6°N (dp/dm = 12.1/24.2°)										

D and *I* are the mean declination and inclination respectively from *n* sample directions derived from a site population of *N*. *R* is the length of the resultant vector, *k* is the Fisher precision parameter (= $(n-1)/(n-R)$) and α_{95} is the radius of the cone of confidence about the mean direction in degrees; dp and dm are the radii of the oval of confidence about the pole position derived from the mean direction in the colatitude direction and at right angles to it, respectively.
* Sites excluded from group mean calculation; see text for explanation; + = overturned beds.

mean direction but is indeterminate according to the classification scheme of McFadden and McElhinny (1990).

The fold and reversal tests suggest that the predominant hematite-resident remanence may predate the ~2900 Ma deformation of these rocks. The corresponding pole position (356°E, 32°N, dp/dm = 12/17°) therefore merits interpretation as a mid-Archaean pole from the Singhbhum–Orissa Craton. Of the six excluded component directions, four have shallow inclinations with NNW and SSE declinations (TH1 (m), K1, KB1 and B1) and provide weak evidence for a dipolar axis of contrasting age recorded in these rocks. The mean of this subsidiary ('B') group (*D*/*I* = 347/3°, see Table 1) is hardly changed by tilt adjustment; the precision is also little affected (k = 13.2 to k = 15.4); there is therefore no evidence for the age of the remanence recorded by these components with respect to deformation, although the corresponding pole position (195°E, 37°N) may be of Late Archaean age (see Section 7).

7. Discussion

The AMS fabrics cannot be directly used to correct the mean remanence direction for the effects of fabric but since the magnitude of AMS is small, it does suggest that any such deflection is likely to be minimal. The most important finding of low temperature susceptibility study of these rocks is the widespread occurrence of small amounts of mul-

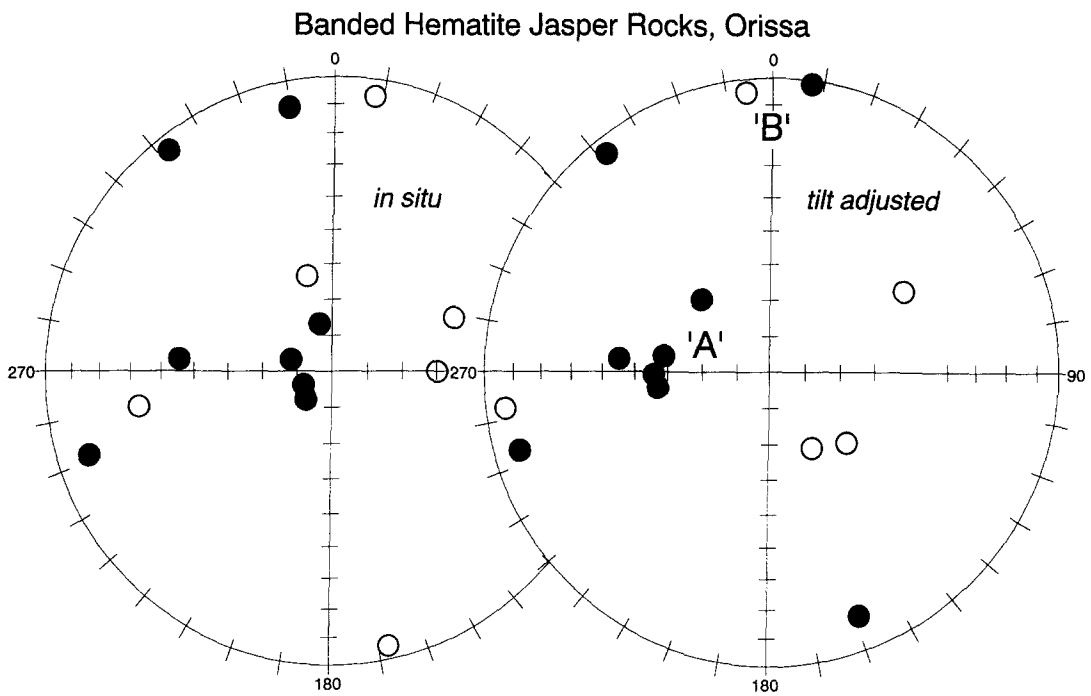


Fig. 6. Component directions isolated at sites in the hematite jasper rocks before and after adjustment for tilt. Equal area projection. Solid symbols are lower hemisphere plots and open symbols are upper hemisphere plots.

tidomain (MD) magnetite (Mallik et al., 1993). This appears to be ubiquitous as a primary phase in the specularite bands but it is unclear whether magnetite is generated in the jaspers during laboratory study. Reduction is not observed in all samples heated to above 600°C and it is suggested that coexistence of magnetite and hematite in the BHJ rocks could be accommodated in three ways: (i) the magnetite is produced by magnetotactic bacteria, (ii) it is formed by reduction of hematite, or (iii) it is of primary detrital origin and has largely been converted to hematite during diagenesis.

In recent years biogenic magnetite has been considered to be an important contributor to the magnetisation in sediments (e.g. Stolz et al., 1990). However, it grows in superparamagnetic (SP) and single domain (SD) grain sizes; furthermore, other work suggests that SD and SP magnetites can be formed abiogenically by oxidation of ferrous iron solutions at near-neutral pH (Maher and Taylor, 1988). This evidence appears to exclude mechanism (i). Mechanism (ii) is also unlikely: although magnetite is produced in the rock magnetic experiments by reduction of hematite, this only happens at temperatures above 500°C and there is no sign that the sampled BHJ rocks were ever subjected to temperatures above 200°C during their long geological history. Hence the magnetite would appear to have a primary detrital origin and the hematite may then be either of primary chemical sedimentary origin and/or have resulted from comprehensive oxidation of a chemical sedimentary magnetite precursor during diagenesis.

It is our preferred interpretation that the fold and reversal tests, whilst statistically inconclusive, may represent a pre-folding magnetisation. This interpretation implies that remanence was fixed before regional folding and emplacement of the Singhbhum Granite at 2900 Ma, and it is supported by lack of correspondence with episodes of younger overprinting in eastern India at ~2600 Ma and ~1000 Ma. The metamorphic complex in this region was also intruded by later granites (Mayurbhanj, Soda, Arkasoni and Chakradharpur Granites) and thick lava sequences were emplaced in adjoining basins (Dhanjori, Simlipal and Dalma) during Palaeoproterozoic times (Sarkar and Saha, 1977; Saha and Ray, 1984). The absence of a coherent remanence representative of these later times suggests that magnetic overprint-

Table 2
Late Archaean–Palaeoproterozoic palaeomagnetic poles from India

Code	Formation	Age	Pole: °E	Pole: °N	Reference
Singhghum and Dharwar Cratons, India:					
BHJ1	Banded Hematite Jasper Rocks, Orissa	3150–2900	356	32	This study
BHJ2	BHJ Facies, 'B' group		320	7	This study
QM1	Quartz–magnetite rocks, Khammano		355	−6	Poornachandra Rao et al. (1989)
SK	Sukinda Chromite Formation	ca. 2600	40	−39	Kumar and Bhalla (1984)
CH	Palni Hills Charnockite	ca. 2600	331	−35	Bhimasankaram (1964)
QM2	Quartz–magnetite rocks, Kistaraopalli	2590	330	48	Subha Rao and Radhakrishnamurty (1979)

ing by these later events was not significant in the BHJ facies.

Palaeomagnetic study of the Precambrian of the Indian subcontinent has been hampered not so much by paucity of data as by lack of good age control on the poles which are available. Thus, numerous studies of dense dyke swarms cutting the metamorphic basement appear to be broadly applicable to Mesoproterozoic times but are of little interpretive value (see Hargraves and Bhalla, 1983; Piper, 1987). In the Singhbhum–Orissa Craton Verma and Prasad (1984) distinguish three groups of dyke directions; they are distinct from directions of this study and two are removed from the Deccan overprints pervasive in many Indian rocks; they appear to be constrained to the interval 1600–950 Ma. Although some of the Indian dykes are assigned to Late Archaean–Palaeoproterozoic times, they are variably altered and it seems likely that the magnetisations are appreciably younger than this (Hargraves and Bhalla, 1983).

Accordingly the compilation of poles in Table 2 excludes the dyke results and includes only data from basement and supracrustal rocks assigned to Late Archaean or Palaeoproterozoic times. Two results from charnockites in the eastern Singhbhum Craton (although consistent with an assigned age of ~2600 Ma on the basis of U–Pb ages) are excluded as probable remagnetisations at ~1000 Ma (McElhinny, 1969 and Dr. T. Radhakrishna, pers. commun.). Thus a polar movement of some 80° during Late Archaean times is recorded between the palaeopole from the banded hematite jasper rocks of this study (BHJ1 in Table 2) and the poles from quartz–magnetite and chromitic rocks in the Singhbhum and Dharwar Cratons assigned to ~2600 Ma. This movement may embrace the subsidiary 'B' Group (Table 2).

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